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An evaluation of thinning to improve habitat for capercaillie (*Tetrao urogallus*)

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ABSTRACT

In Scots pine (*Pinus sylvestris*) forests the composition of the ground flora can be affected by the amount of light reaching the forest floor, influencing the balance between the three common ericaceous shrubs bilberry (*Vaccinium myrtillus*), cowberry (*Vaccinium vitis-idaea*) and heather (*Calluna vulgaris*). A pinewood ground flora with more than 20% bilberry cover is considered good habitat for capercaillie (*Tetrao urogallus*), a large forest grouse of considerable conservation interest throughout Europe. Old, semi-natural Scots pine woodland is considered its prime habitat in Scotland, although this is limited in area compared to twentieth century planted forests. Action to manipulate environmental conditions within Scots pine plantations by altering light levels to favour bilberry through thinning and felling could potentially increase greatly the area of available capercaillie habitat in Scotland. We implemented a five year study to look at bilberry response to variable intensity thinning in two Scots pine plantations, where thinning followed management guidance for capercaillie available at the time. Bilberry was present in both forests but a positive response to thinning was not universal; although at both sites bilberry cover increased significantly over five years with levels >20% cover reached, this could only be attributed to the thinning treatment at one of the sites. A treatment of small patch clearfelling did not lead to losses in bilberry. Management guidance published after the trial had begun, identified the appropriate intensity of thinning for enhancing bilberry cover at our study sites, indicated by the relative increase of bilberry in the plots where the prescription had been followed. Although there was no significant treatment effect by year five, 42% average bilberry cover was reached at one site tested. However the format of this guidance, a range of stem density-tree height combinations, was difficult to apply using typical forest management data and we explore redefining the guidance as a post thinning stand basal area range. We suggest >22 to <31 $m^2\,ha^{-1}$ basal area would be appropriate in Scots pine plantations established at normal spacing and subject to the commonest form of selective thinning regime. This range in basal area can be achieved without conflicting with management for timber production. Our results also support small patch clearfelling as a method of diversifying plantation age structure which is compatible with maintaining capercaillie brood habitat.

1. Introduction

Capercaillie (*Tetrao urogallus*) is a bird of boreal and temperate forests of north and central Asia and Europe, but its range has greatly reduced in western and central Europe as a result of many local extinctions (Storch, 2001; Watson and Moss, 2008). Conservation concerns prompted by these declines have resulted in capercaillie being included in Annex I, II(B) and III(B) of the EC Birds Directive (Directive 2009/147/EC). In addition, under the EC Birds and EC Habitats Directive (Directive 92/43/EEC), capercaillie is a specific designated feature of many of the Natura 2000 sites, and large parts of the capercaillie range within the EU countries of central and western Europe are covered by Natura 2000 designated sites. The responsibility for protecting the interests of these sites is shared between different authorities, including state forest agencies who have implemented policy changes to integrate forestry practices with capercaillie habitat requirements as a key conservation measure (Eurosite, 2007; Storch, 2007, 2001; The Scottish Government, 2012).

In the UK, capercaillie became extinct in the eighteenth century but was re-introduced successfully during the first half of the twentieth century. However, having suffered declines in the last 25 years, the capercaillie population is now limited to certain areas of Scotland (Eaton et al., 2007; Wilkinson et al., 2002), with the most recent population estimate of 1285 (95% CI: 822–1882)

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individuals (Ewing et al., 2012). Old, semi-natural Scots pine woodland has been considered the prime habitat of capercaillie in Scotland and contain the highest densities of birds (e.g. Picozzi et al., 1992) although this could be confounded by the higher probability of detecting birds in native pinewoods compared to plantation woodlands (Ewing et al., 2012). Recent surveys confirm conifer plantations, especially Scots pine, can support capercaillie populations (Petty, 2000; Eaton et al., 2007). Semi-natural Scots pine woodlands in Scotland cover less than 20 thousand hectares (Forestry Commission, 1999) but Scots pine plantations potentially offer at least five times more area of habitat (Mason et al., 2004). Extensive new plantings of Scots pine woodland have occurred in the last 20 years in Scotland (Forestry Commission, 2013, 2002; Anon, 2002) and these provide large opportunities for capercaillie habitats in the future. An important way of halting species decline in Britain is foreseen by enhancement of capercaillie productivity (Caledonian Partnership, 2002) and specific actions have been taken including improvements to the condition and extent of brood habitat to achieve this (Moss et al., 2000; Storch, 2001).

Woodlands provide a number of habitat factors required by capercaillie. For habitats used in winter, for example, structural and compositional features such as the presence of pine and spruce trees and solitary, branched trees have been shown to be important (Gjerde, 1991a; Storch, 2001; Bollmann et al., 2005). Whereas a good spring and summer habitat, providing both cover and food for capercaillie adults and broods, is considered to be a wood with a developed ground flora that is rich in bilberry (*Vaccinium myrtillus*) (Picozzi et al., 1999; Storch, 2001, 1993). Chicks in their first 3–4 weeks are almost entirely dependent on invertebrates; in Scotland primarily Lepidoptera larvae but also ants, spiders and beetles (Picozzi et al., 1999). Later, the birds feed on the leaves and berries of bilberry (Summers et al., 2004; Spidø and Steun, 1988). The extent of cover of bilberry has been shown to be positively correlated with the biomass of invertebrates (Lakka and Kouki, 2009; Summers et al., 2004) and positively related to breeding success of capercaille (Baines et al., 2004). Recent work by Hancock et al. (2011) found that management which increased bilberry cover led to more capercaillie usage and more chick food.

Ground vegetation is strongly influenced by the canopy, which in turn affects the levels of usually several different site resources governing composition of the ground flora (Wagner et al., 2011). However, previous studies suggest that within woodlands on the same, poor and acidic soil type e.g. pinewoods, the balance of the ericoid species including bilberry and shade tolerant grasses such as wavy hair-grass (*Deschampsia flexuosa*), is controlled primarily by the availability of light (Humphrey, 1996; Hester et al., 1991). However, herbivores also influence bilberry cover and depending on the context, have been shown to have positive (Hancock et al., 2010) or negative effects (Welch, 1998). Action to manipulate environmental conditions within plantations offers the potential for increasing brood habitat in particular by altering light levels to favour bilberry (Kortland, 2006).

The relationship between stand structure and ground flora composition has been described from space-for-time substitution studies (multiple sites of different ages studied in same year) in semi-natural Scots pine woods and plantations of other conifer species (Picozzi et al., 1992) and a guide to quantify capercaillie habitat and determine appropriate stand management (target stem density per hectare and tree height) has been produced for Scotland (Moss and Picozzi, 1994). Parlane et al. (2006) reported findings relating bilberry cover to canopy light transmittance in Scots pine stands in guidance published subsequent to the commencement of our trial. However, as both these studies were correlative and focussed on management to benefit capercaillie, application of the resulting conservation guidance to management of Scots pine plantations is restricted in two ways. Firstly, there is the issue of the transferability of the relationships to stands of different origin, structure and stand history and whether the recommended practice is indeed effective. This can be addressed by testing experimentally whether correlations found in these earlier studies reflect a causal link, as our work described in this paper has attempted to do. Secondly, stand management rarely occurs solely to deliver capercaillie habitat, and habitat management may need to be achieved by modifying operations primarily conducted for other reasons such as timber production. Regular thinning is recommended to maximise timber production (Rollinson, 1988) and timber quality of Scots pine plantations (MacDonald et al., 2010a) following prescriptions given in yield models for forest management (Edwards and Christie, 1981; Matthews, 2008). However, as indicated in the National Inventory of Woodlands and Trees 1995–1999 (Quine et al., 2007; J Gilbert pers comm., 2007), between 55% and 80% of the plantation forests in Scotland remain unthinned possibly due to the low value of the produce in early thinnings and the net cost of the operations (Mason, 2007). Forest management is influenced by timber prices, with more money for forestry activities being available when the timber prices are high (Mason (2007). Timber prices showed a 70% decline between 1995 and 2003 with a varying amount of recovery since 2006 (IPD, 2004, 2011). In contrast, management recommendations for capercaillie conservation, including glade creation and thinning within plantation woodlands (Moss and Picozzi, 1994; The Capercaillie LIFE Project, 2004) could result in overcutting and early, intensive thinning, with the associated losses in volume production (Rollinson, 1988) and impacts on the timber quality (Ikonen et al., 2009; MacDonald et al., 2010b; Mäkinen and Isomäki, 2004). Providing appropriate guidance and incentives for forest management in capercaillie areas in Britain which can balance timber production and conservation objectives is therefore a recognised need.

In 2002, EU funding was available to encourage forest management interventions to improve capercaillie habitat in pinewoods in the oceanic boreal region (The Capercaillie LIFE Project, 2004). This paper reports on the success of the stand thinning measures (and specifically variable intensity thinning and small glade creation) in improving spring and summer habitat quality for capercaillie broods and adults. These measures were assessed in a designed study. The main aims of this study were:

1. To test whether the cover of bilberry increased in response to variable intensity thinning and small patch clearfelling.
2. To assess the applicability of two different sets of current conservation management guidance (Moss and Picozzi (1994) and Parlane et al. (2006)) to Scots pine plantation management, and thereby strengthen the evidence base for this conservation measure (Sutherland, 2006).

2. Methods

2.1. Study sites

Study sites were located in two forests (Inshriach and Novar) in northern Scotland, in areas identified during the EU LIFE Nature project as typical Scots pine plantations requiring thinning to improve brood habitat (Kortland, pers. comm.). The sites have a similar climate, experiencing around 900 day-degrees above 5 °C and a summer and winter rainfall of 400 and 550 mm, respectively (Pyatt et al., 2001). Mean (minimum and maximum) temperatures are 0.0 °C and 5.2 °C in January and 9.4 and 16.1 in July (Met Office, 2013). Both sites have similar podzolic soils. (Table 1). Iron and humus-iron podzols are the main soil type associated with pinewoods in the oceanic boreal region (Wilson and Puri, 2001; Godbold and Lukac, 2011).

Table 1
Summary of site and stand information for the two study sites (Inshriach and Novar).

Study site	Inshriach		Novar	
Location	Strathspey, Highland, Scotland 57°7′N, 3°51′W		Easter Ross, Highland, Scotland 57°46′N, 4°27′W	
Study site area	50 ha		30 ha	
Aspect	NW		W–SW	
Slope	4.5°		12.5°	
Planting year of Scots pine stand	1961		1962	
Average stem density	1700 stems ha^{-1}		1600 stems ha^{-1}	
Average diameter at breast height(1.3 m)	16 cm		18 cm	
Estimated yield class[a] (Site index[b])	6–8 (18–25)		8 (22–25)	
Soil type [c]	Podzol		Podzol	
Typical profile [d]	O	0–3 cm	O	0–6 cm
	Ah	3–7 cm	Ah	6–12 cm
	E	7–20 cm	E	12–25 cm
	Bhs	20 – 60+	B(g)hs	25–50 cm
			Bx	50 cm+
	Stones and Gravel 50%		Stones 20%	
pH (H20) (1:2.5) [e]				
A Horizon – average (minimum–maximum)	4.26 (3.75–4.78)		4.27 (4.08–4.37)	
B Horizon – average (minimum–maximum)	4.89 (4.45–5.16)		4.9 (4.77–5.01)	
Ground flora-vascular plant species of frequent occurrence (average %cover)	Bilberry *Vaccinium myrtillus* (18%) Wavy hair-grass *Deschampsia flexuosa* (40%) Cowberry *V. vitis-idaea* (6%) Heather *Calluna vulgaris* (6%)		Bilberry (6%) Wavy hair-grass (40%)	
Ground flora-moss species of frequent occurrence	Glittering wood-moss (*Hylocomium splendens*) Red-stemmed feather-moss (*Pleurozium schreberi*)		Glittering wood-moss, Red-stemmed feather-moss Waved silk-moss (*Plagiothecium undulatum*)	

[a] Estimated yield class follows Edwards and Christie (1981).
[b] Site index after Hägglund and Lundmark (1977).
[c] Classification according to the standard Forestry Commission scheme (Pyatt and Suarez, 1997).
[d] Horizon nomenclature follows Avery (1990).
[e] pH was measured using a glass-electrode metre in a 1:2.5 w/v suspension of soil in de-ionised water.

The plantations at the Inshriach and Novar sites were established in the early 1960s. At the start of the study the stands were stocked to a similar density, with trees of similar average girth (Table 1). The management history of these two sites was however different. Inshriach Forest is managed by the government forestry agency, the Forestry Commission, and had already received its first thinning. Novar Estate is a privately owned estate; at the commencement of study the stand had not yet been thinned and no previous crop records were available. Although the composition of ground vegetation was similar and bilberry occurred frequently at both sites (Table 1), at the start of the study, Inshriach had a higher cover of bilberry (just under 20% cover) compared to Novar (less than 10% cover). Mostly roe deer (*Capreolus capreolus*) but also red deer (*Cervus elaphus*) are present at both sites.

During the course of the study (2002–2007), both sites experienced a wet growing season in 2007 but below average rainfall in 2003 and 2006; temperatures were above average for both sites with the growing season of 2003 being particularly warm (Met Office, 2010; Stathspey Weather, 2010).

2.2. Thinning treatments

Thinning treatments were designed and applied with the aim of enhancing the habitat for capercaillie. In line with the established guidance for enhancing bilberry cover (Moss and Picozzi, 1994), the proposed variable thinning treatment was intended to be conducted at normal to twice the normal thinning intensity suggested in yield models for Scots pine (Edwards and Christie, 1981). A patch clearfell treatment was included at Inschriach, consistent with recommendations for glade creation favourable for broods, and a no thin control was included at both sites.

At Inshriach, the original study layout was seven randomised blocks of four 50 m × 50 m plots. Four treatments (Control, Normal Thin, Double Thin and clearfell) were randomly allocated to plots within a block. However, application of the treatments in the course of normal harvesting operations did not maintain a consistent separation between thinning intensities and so both were amalgamated into one Variable Thin treatment for analysis purposes; in one block a proposed thinning was not carried out on one plot. The final layout comprised 28 plots (six blocks with one Control plot, one clearfell plot and two Variable Thin plots, and one block with two Control plots, one clearfell plot and one Variable Thin plot) (Fig. 1).

At Novar the study was laid out on a site that had been very recently thinned to benefit capercaillie habitat under the auspices of the LIFE project. The operations aimed to create a variable thinned woodland by alternating strips of variable thinning with non-thinned areas; the strips were arranged systematically along a forest road and not with respect to existing site conditions. The clearfell treatment was not applied at Novar in order to maintain timber production potential for the owner. Eight 50 m × 50 m plots were randomly allocated to the thinned and non-thinned areas, to establish four plots with a Control treatment and four with a Variable Thin treatment (Fig. 1).

Thinning and felling was carried out mechanically by a wheeled harvesting machine, access routes or racks were cut in the forest and trees were removed from the plots by the harvester stationed on the racks; the harvester therefore did not enter and impact on the ground conditions of the plot. The thinning residues (branches and foliage from the felled trees) were removed from the plots. These were used in the construction of brash mats in the racks for the harvester to move over, thereby protecting the soil from compaction and erosion.

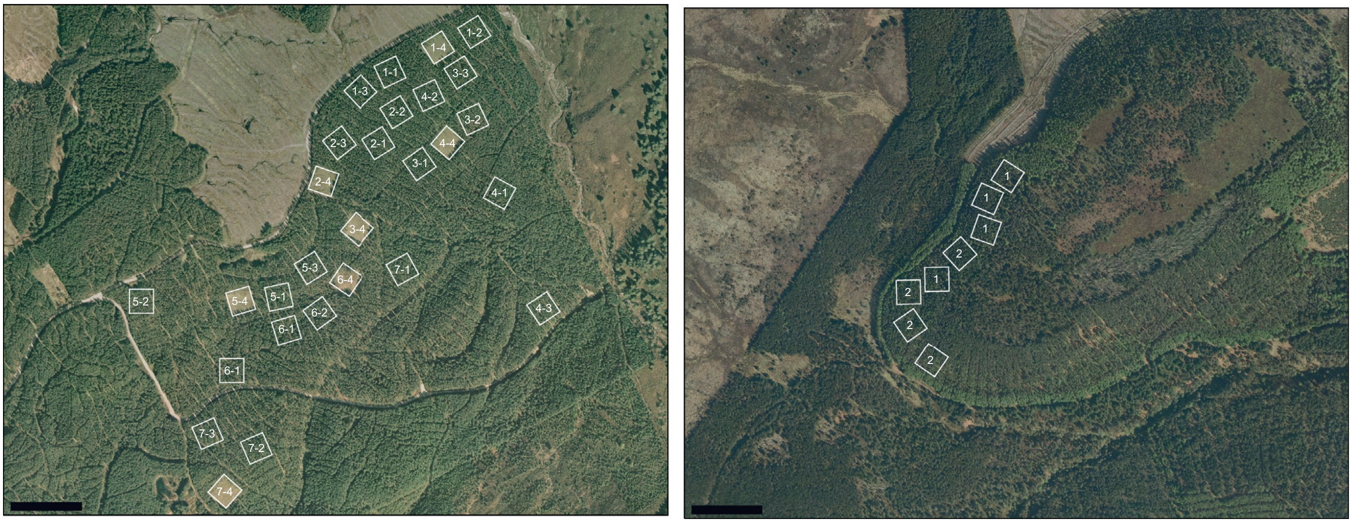

Fig. 1. Plan of study sites with locations of 50 m × 50 m treatment plots (left) Inshriach site with each plot labelled by block number (1 to 7) and treatment (1 = Control, 2 = Variable Thin (planned to be at normal thinning intensity), 3 = Variable Thin (planned to be at twice normal thinning intensity), 4 = clearfell); (right) Novar site with plots labelled by treatment (1 = Control, 2 = Variable Thin), no blocking was applied.

For both sites, pre and post thinning basal area ($m^2 ha^{-1}$) and stem density (stems ha^{-1}) in each plot was measured shortly following intervention from cut stumps and standing trees; stem taper was assessed in each plot to allow conversion of cut stump diameter to diameter at 1.3 m (conventionally used to calculate basal area) (Matthews and Mackie, 2006). Measures of stand top height (m), consistent with the methods of Matthews and Mackie, 2006, had been taken by the site managers prior to thinning at Inshriach only.

2.3. Measuring herbivore effects

Bilberry responses to thinning could be confounded by grazing by herbivores. In an attempt to separate grazing from thinning effects, one of the vegetation assessment quadrats was protected from herbivores (see Section 2.4.). As a further measure, presence of dung of a range of herbivores (roe deer, red deer, sheep (*Ovis aries*) and rabbit (*Oryctolagus cuniculus*)) (MacDonald and Barrett, 2005) was assessed annually in each of the circular quadrats (see Section 2.4). At the same time, any droppings of capercallie were also recorded. However, too few records of herbivore dung or capercaillie droppings were collected during the trial for analysis.

2.4. Vegetation assessments

At both sites, the vegetation assessments followed the same split plot design; each 50 m × 50 m plot contained five, 2 m × 2 m square (vegetation assessment) quadrats, one of which was caged against herbivores. Each square quadrat was positioned at the centre of a circular quadrat (5.6 m radius) used for assessment of stand basal area (Matthews and Mackie, 2006) (see Fig. 2).

Pre-thinning baseline ground vegetation assessments were made within the same year (growing season) that the thinning was applied (July 2003 (year 1) at Inshriach and September 2003 at Novar) and at the end of the next four growing seasons (September to October) in 2004 (year 2), 2005 (year 3), 2006 (year 4) and 2007 (year 5). Two sets of recorders were used for the duration of the trial, the first assessed both sites in year one and two, and the second both sites for the remainder of the study. All observers were trained to the same standard prior to taking part in the study. Bilberry was in leaf at each assessment time. Although levels of defoliation by caterpillars can vary from year to year (Atlegrim and Sjoberg, 1995; Hancock, 2011), we assumed these were similar in areas under different treatments. Percentage cover of all ground and moss layer vegetation species, with cover of greater than five percent, was recorded and cover was estimated to the nearest 5%. The cover of species occurring in different layers within the vegetation was assessed separately giving a total cover value for the quadrat; this value could therefore exceed 100%.

2.5. Data analysis

2.5.1. Bilberry response to treatments

We used SAS version 9.3 for statistical analysis (SAS Inst., 2000). We tested the response of bilberry (absolute% cover values) to treatment, caging, year and their interactions in a repeated measures analysis of variance (ANOVA rm), using a linear mixed effects model, REML (Patterson and Thompson, 1971), to incorporate the random block, plot, and quadrat effects and fixed treatment effects (treatment, caging and year) and to allow for any imbalance in design, as appropriate. Covariance structures were described using an autoregressive function which assumes decay in correlation with increasing separation between measurements (i.e. time). Data were analysed at the quadrat (sub-plot) level to utilise the split plot design i.e. to include analysis of treatment, caging and their interaction. Examination of residuals suggested no data transformation was required.

2.5.2. Comparison of response with expectations based on existing guidance

In order to test the suitability of existing guidance, we contrasted our measurements with the published management guidance of Moss and Picozzi (1994) and Parlane et al. (2006). We determined the expected response by locating our stands within the guidance frameworks using stand measurements and published yield models. We made two comparisons:

i. To hypothesise whether the basal area and stand conditions of the study sites would be suitably changed by thinning to create the conditions to encourage bilberry cover, [Hypothesised response]. This comparison was carried out in order to assess whether the thinning regime used at the study sites

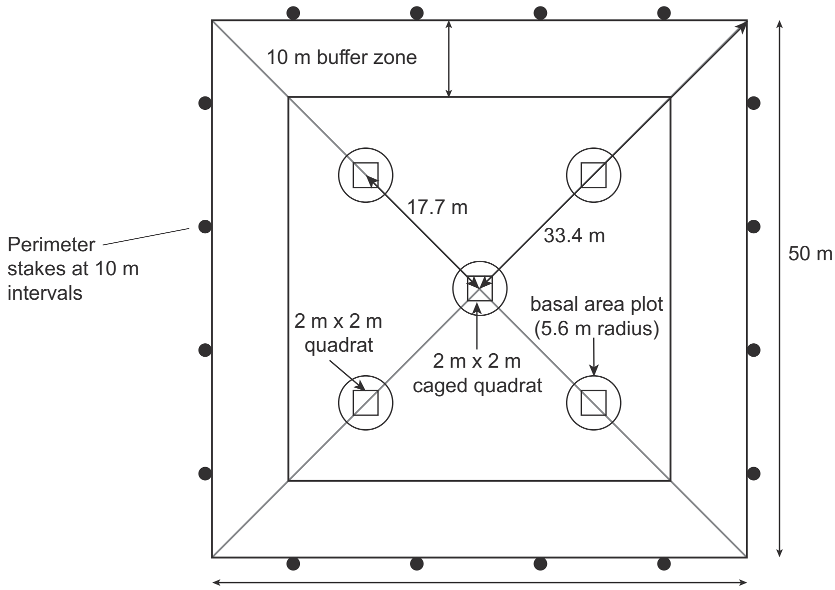

Fig. 2. Plot layout used at both study sites (Inshriach and Novar) showing area thinned/clearfelled (for treatment plots) or left unthinned (for control plots), and locations of basal area (round) and vegetation (square) assessment quadrats.

followed that given in the published management guidance and to indicate whether the management guidance was suitable for use in plantation situations.

ii. To determine whether at the end of the trial the change in bilberry cover at our study sites over the trial period was better where prescriptions from the guidance had been followed given the post thinning stand conditions (stem density and tree height), [End comparisons].

For the ‘Hypothesised response’ we plotted the stem densities and stand heights at the intervention times in a model Scots pine plantation of the same Yield Classes (YCs) as our study sites. We compared the guide stem densities and tree heights considered to benefit bilberry given by Moss and Picozzi (1994) and Parlane et al. (2006) and our trial stands according to the stem densities we planned to create following a thinning intervention of between normal and twice normal thinning intensity.

For the ‘End’ comparison we compared stem density and tree height data for each plot to those given in the thinning guidelines proposed by Moss and Picozzi (1994); Fig. 4, page 11) and by Parlane et al., (2006; Fig. 2, page 277). This could be completed reliably for the plots at Inshriach, as only here the assessment of tree height had followed appropriate methods (Matthews and Mackie, 2006). For Inshriach data, we allocated plots to three treatments according to whether they had been thinned to prescription (T), had received ‘other’ thinning treatments (O) by either being under-thinned (post thinning stem density greater than that prescribed) or over thinned, this later category included clearfelling of plots, or were unthinned control plots (C). We analysed the response of bilberry to the three treatments over the five years of the study, following the method described in Section 2.4.1.

As a further comparison with the management recommendations of Parlane et al. (2006), we derived the stand basal area (in $m^2\ ha^{-1}$) associated with the prescribed stem density and tree heights in two ways. Firstly, using the canopy transmittance (the proportion of incident radiation that is transmitted through the canopy) equation (Parlane et al., 2006) and the relationship between canopy transmittance and stand basal area (in $m^2\ ha^{-1}$) derived for Scots pine stands (Hale et al., 2009). Secondly, using the yield models for Scots pine for stands established at normal (2 m) initial spacing and subject to the commonest type of selective thinning (‘intermediate’ thinning) (Matthews, 2008), and of Yield Class 4–14. This range of Yield Class represents the variation in growth rates which can be sustained by Scots pine stands within the distribution range of capercaillie in Scotland (Forestry Commission, 2011) on sites where the soil fertility is suitable for bilberry growth (MacDonald et al., unpublished).

3. Results

3.1. Bilberry response to thinning treatments

At the start of the trial before the thinning, basal area within the different treatment plots were similar between the two sites except for the clearfell (CFELL) treatment at Inshriach where the pre-thinning basal area was lower (mean of 27 $m^2\ ha^{-1}$) (Table 2). Thinning reduced the basal area below that of the control in all treatments. The mean post-thinning basal areas in the variably thinned were similar at both sites and the transmittance, as derived from the basal area measurements (Hale et al., 2009), was 0.35 at Novar and 0.33 at Inshriach (Table 2).

The change in bilberry cover over time was generally positive. The cover of bilberry increased over the five years at both of the study sites but the influence of thinning treatment was only apparent at one site (Novar). At Inshriach, the only significant effect was that of year (ANOVA rm, $F_{4,474} = 16.45$, $P < 0.0001$) with mean %cover of bilberry increasing from 16% in year one to 32% in year five (Fig. 3a). At Novar, the increase in bilberry cover was larger in the variably thinned plots than the control plots (treatment*year, ANOVA rm, $F_{4,140} = 4.54$, $P = 0.0018$), resulting in a mean %cover of bilberry of 24% in the variably thinned plots and 5% in the control plots by year five (Fig. 3b). Year effect was significant (ANOVA rm, $F_{4,140} = 8.65$, $P < 0.0001$) but treatment effect alone was not. There was no effect of caging on bilberry cover at either site.

The clearfell treatment was only applied at Inshriach. A reduction in bilberry cover in response to clearfell treatment was not shown by the results (Fig. 3a) as no significant treatment effect (or treatment*year effect) was detected. The response of bilberry cover to clearfell treatment was therefore not different from that seen in the Control or the Variable Thin treatment.

Table 2
Mean values (S.E.) for measures of stand variables within the plots at the two study sites before and after thinning treatments (control (CTRL), variable thin (VT), clearfell (CFELL)).

		Inshriach			Novar	
		CTRL	VTHIN	CFELL	CTRL	VTHIN
Pre-thinning	Basal area average (m^2 ha^{-1})	35 (2.0)	37 (1.0)	27 (1.0)	39 (3.6)	35 (1.8)
	Transmittance[a]	0.28 (0.02)	0.26 (0.01)	0.35 (0.01)	0.25 (0.03)	0.28 (0.02)
Post-thinning	Basal area average (m^2 ha^{-1})	35 (2.0)	29 (0.9)	0	39 (3.6)	27 (1.6)
	Transmittance[a]	0.28 (0.02)	0.33 (0.01)	0.78	0.25 (0.03)	0.35 (0.02)
	Post thin stem density average(stems ha^{-1})	1673 (203)	1089 (74)	0	1690 (123)	920 (41)

[a] Derived from basal area measurements using transmittance and stand basal area relationship (Hale et al., 2009).

3.2. Comparison with conservation management thinning guidance

3.2.1. 'Hypothesised response'

It appeared that the thinning regime applied at the study sites would match Moss and Picozzi (1994) management guidance but only partially follow Parlane et al. (2006) guidance. We forecast (i) a response likely to result in the bilberry cover given by Moss and Picozzi (1994) in all plots at Inshriach (Fig. 4a) and all but the more lightly thinned plots at Novar (on the basis that Novar is represented by a YC8 stand subject to a 10 year delay to first thinning) (Fig. 4b) and (ii) based on Parlane et al. (2006) guidance, the creation of conditions optimal for bilberry in the plots subject to normal thinning at Inshriach in the more productive compartments (YC8- second intermediate thinning) (Fig. 4a), and in the plots most intensively thinned at Novar (Fig. 4b).

3.2.2. 'End' comparisons

For Inshriach, on the basis of the Moss and Picozzi (1994) management guidelines, all but two of the 13 variably thinned plots were classed as having been thinned to prescription (T), the remaining variably thinned and the seven clearfell plots were classed as other thinning (O). Following Parlane et al. (2006) guidelines, six of the variably thinned plots at Inshriach were classed as thinned to prescription (T), with all remaining thinned and clearfelled plots being classed as 'other' thinning.

The ANOVA rm analysis of bilberry cover in the three treatments (T, O, C) over five years showed that whilst treatments diverged significantly over time (treatment*year, ANOVA rm, $F_{8,476} = 3.03$, $P = 0.0025$) with prescription thinning (according to Parlane et al. (2006)) tending to give a rising mean value of bilberry cover compared to other treatments, the final year means did not differ significantly between treatment (treatment ANOVA rm, $F_{2,20.6} = 0.57$, $P = 0.5745$), although treatment T had the highest predicted cover of bilberry (42%) (Fig. 5). There was also a significant year affect (ANOVA rm, $F_{4,476} = 16.89$, $P < 0.0001$). For plots thinned according to the Moss and Picozzi (1994) prescription, there was no significant treatment effect. There was no effect of caging on bilberry cover in either analysis.

Regarding stand basal areas derived from Parlane et al. (2006), a stand basal area between 23 and 26 m^2 ha^{-1} would appear to create optimal transmittance (0.37–0.33) identified for bilberry. However, according to the yield models for typical Scots pine stands, Parlane et al. (2006) prescribed stem density and tree height combinations are associated with basal areas captured in the range of between >20 and <31 m^2 ha^{-1}. 'Typical' model Scots pine stands are considered those established at the normal (2 m) spacing and which have been subject to the commonest type of selective thinning ('intermediate' thinning) regime (Matthews, 2008).

Although not formally tested, it would appear that almost all the plots at Novar were thinned in line with the Parlane et al. (2006) guidance. Further, the basal area of these plots ranged from 23 to 29 m^2 ha^{-1} (mean 27 m^2 ha^{-1}; transmittance 0.35 (Table 2)) close to optimal range (based on stand conditions and transmittance relationships) and within the broader range (based on stand conditions and yield models) indicated by Parlane et al. (2006). In contrast, of the 13 variably thinned plots at Inshriach (mean basal area 29 m^2 ha^{-1}; transmittance 0.33 (Table 2)), eight plots achieved post thinning basal areas within the broader range, with the remainder having basal areas greater than that indicated by the Parlane et al. (2006) prescription.

4. Discussion

4.1. Bilberry response to thinning and clearfell treatments in study Scots pine plantations

We confirmed that appropriate bilberry cover for capercaillie can be obtained in Scots pine plantations. The final cover values of bilberry exceed 20% in all but the Control treatment at Novar. In the context of the study, this can be regarded as a successful outcome as more than 15–20% bilberry cover is considered suitable for capercaillie breeding success (Baines et al., 2004; Storch, 1994; Summers et al., 2004). Our study provides evidence that applying variable intensity thinning to Scots pine plantations can improve capercaillie brood habitat, however, bilberry cover only responded to treatment in one but not both of the Scots pine plantations studied. Further, the results of the trial only partially support the use of the variable thinning technique as described in management guidance (Moss and Picozzi, 1994; Parlane et al., 2006), to encourage bilberry.

Lack of, or failure to detect treatment effects on bilberry at both sites may be due to differences between the sites and in particular their management history. Although the sites were similar in most respects, Inshriach had been thinned previously and had a higher starting cover of bilberry than Novar. Due to the similarity of the sites with respect to the soil type and in being first rotation plantations established with a monoculture of the same species, we consider that ground flora composition is following the same trajectory of change in response to canopy change at both sites (Hester et al., 1991; Hedwall et al., 2013). However sites may be at different points along the trajectory, with Novar displaying the response to an initial thinning resulting in clearer differences in bilberry cover in the treatment compared to the control. Whereas the increase in bilberry cover we saw in the control plots at Inshriach is likely be a legacy of the previous thinning, and occurring in response to these earlier modifications to the canopy structure that were no longer apparent in our measurements (Thomas et al., 1999). Reduction in bilberry in favour of other competing species may not always be expected to follow canopy regrowth (Strengbom et al., 2004). Nevertheless, we expected that further thinning would provide better conditions for bilberry than in stands where the canopy cover was already recovering and closing (Moola and Mallik, 1998). Bilberry is more abundant in boreal forests where the canopy is neither very dense nor very sparse (Kardell, 1980) and reported occurrence of bilberry under different levels of canopy transmittance in Scots pine woodlands in Scotland, indicates

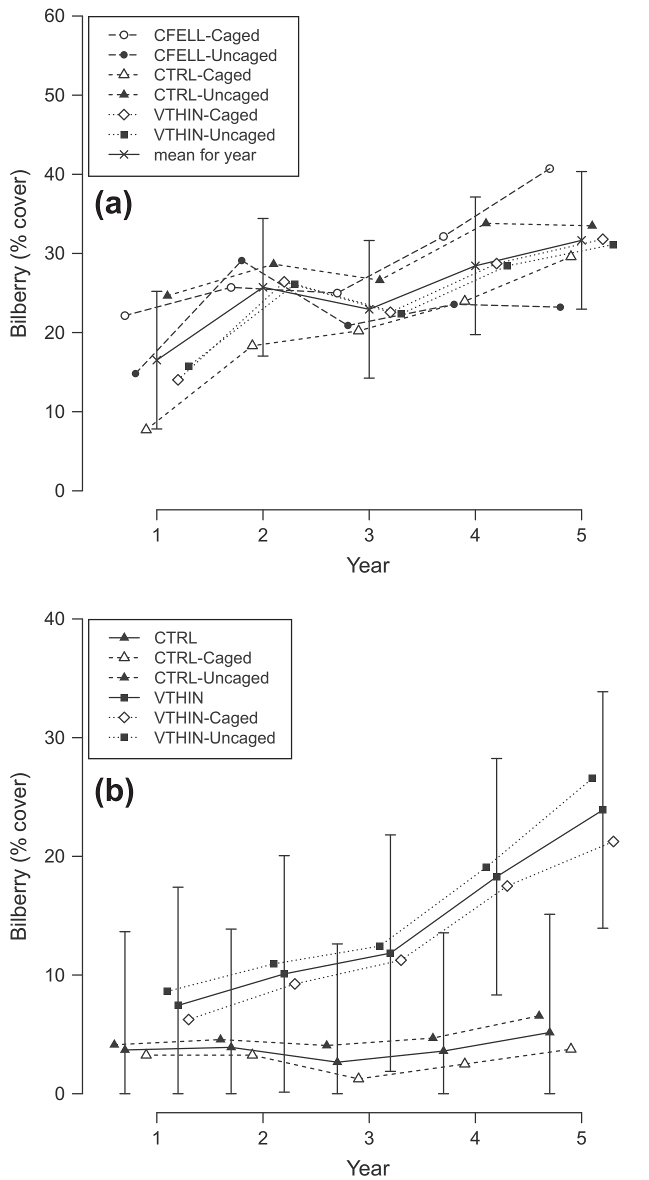

Fig. 3. Bilberry response to variable intensity thinning (VTHIN), to clearfell treatment (CFELL) (one site) and to control (CTRL), with and without caging, at two sites: (a) Inshriach and (b) Novar. Data points for the same year have been offset for ease of viewing.

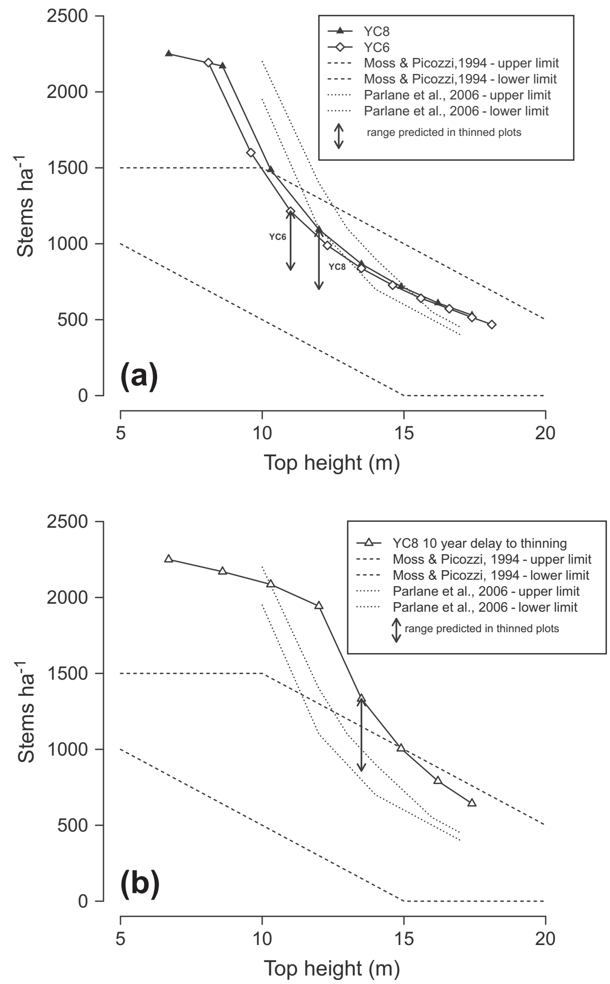

Fig. 4. Stem density and tree heights in a model Scots pine plantations of Yield Class 8 and Yield Class 6, established at 2 m spacing and subject to intermediate thinning treatments; ranges suitable for bilberry (after Moss and Picozzi (1994) and Parlane et al. (2006)) with predicted effects on stem density of normal and twice normal intensity thinning at (a) second thinning as at Inshriach and (b) delayed (by 10 years) thinning as at Novar.

that a relatively narrow range of transmittance provides most favourable conditions (Parlane et al., 2006). It is possible that variable thinning resulted in a broader range of basal areas at Inshriach compared to Novar, and conditions that were too shady for bilberry were created in some of the plots masking a clear treatment response. Given the difference in the starting cover of bilberry at the two sites (average %cover of 18% at Inshriach and 6% at Novar), it may also be important to thin closer to the optimal level where bilberry cover is already high, to gain any benefit from the thinning treatment.

Small patch clearfelling did not appear to be a suitable alternative method to variable thinning, as response of bilberry cover to clearfell treatment was no better than in the Control. Clearfelling is not an intervention previously recommended for enhancing bilberry cover in Scots pine woodlands and a general decrease in bilberry cover with thinning intensity has been reported from a number of studies (Atlegrim and Sjöberg, 1996a; Bergstedt and Milberg, 2001; Crouch, 1986). Five years is a short period of study for bilberry which has been shown to be relatively slow to respond to treatments in other studies (Humphrey, 1996; Bergstedt and Milberg, 2001). This may be why we did not see the large losses in bilberry cover reported by other researchers (Kardell, 1980; Lakka and Kouki, 2009) suggesting that our small patch clearance may have had less of an impact on capercaillie brood habitat than large scale patch clearfell. Questions remain over the overall quality of the brood habitat produced in these open areas as reduced abundance of larvae, linked to the higher phenolic concentration in bilberry leaves, has been reported following clearfelling (Atlegrim and Sjöberg, 1996a,b; Lakka and Kouki, 2009; Nybakken et al., 2013).

4.2. Management guidance for thinning

We have concerns over the application of the current conservation management guidance to plantation management. The management recommendations based on target stem density-tree height combinations available prior to our study (i.e. Moss and Picozzi, 1994) were met in the variably thinned plots at both sites in our study and yet did not appear to define the range of stand conditions which had low enough stand density and were light enough for enhancing bilberry growth.

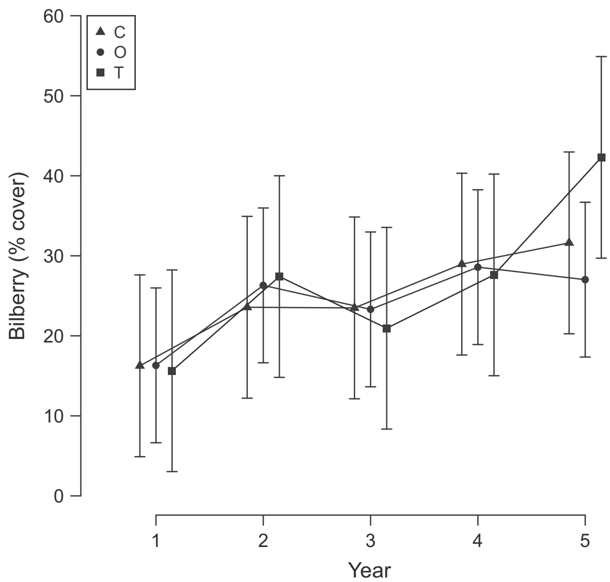

Fig. 5. Bilberry response at Inshriach to thinning applied according to Parlane et al. (2006) prescription (T), not to prescription (O) and to control (C); predicted mean bilberry cover for treatments, with 95% confidence intervals shown. Data points for the same year have been offset for ease of viewing.

The subsequently published and more detailed management recommendations (Parlane et al., 2006), indicated that not all our plots would show a response in bilberry. However, the End comparison indicated that over five years, amount of bilberry cover diverged in the different treatments, with relatively more bilberry in plots which had been thinned according to prescription. This did not result in a difference in bilberry cover at the end of the trial between plots thinned or not to prescription, but we suspect a longer study might have led to a larger final year difference. Although these management recommendations had appeared suitable for use in a plantation setting and fit well with the stem densities of habitat associated with capercallie use in other studies (Gjerde, 1991a,b), we found them difficult to apply. Stem density assessments used by Parlane et al. (2006) differed from the standard forestry methods (Matthews and Mackie, 2006), and despite attempts to align our data, stem densities for a given tree height at our plantation site were still very unlike those studied by Parlane et al. (2006). For example, the 14 m tall stands sampled in their study had 80% less and 12 m tall stands 70% more stems per hectare than we encountered in stands of the same height at Inshriach.

It may be more useful to re-define the Parlane et al. (2006) prescription as target basal area range. As a description of stand characteristics, basal area ($m^2\ ha^{-1}$) is regularly used in forestry. Basal area could be considered a more robust measure in respect of predicting light levels, as a single value can be achieved by various combinations of tree heights and stem densities (Matthews, 2008). For this reason, the accuracy by which target thinning basal area are achieved in practice, is more precise than for stem density targets which are expected to be within 20% of the target in stands at stem densities of c.1500 stems/ha or more to 10% in stands of c.1000 stems ha^{-1} (Mason, pers. comm.). Further, in Scots pine stands, a basal area can be related to a canopy transmittance figure (Hale et al., 2009). For the optimal light levels of 0.37 and 0.33 transmittance (Parlane et al., 2006), a basal area range of 23–26 $m^2\ ha^{-1}$ would be needed (Hale et al., 2009), but suitable stand conditions might also develop in Scots pine plantations which have been subject to intermediate thinning regime when they attain basal areas in the range of >20 and <31 $m^2\ ha^{-1}$ (Matthews, 2008) (transmittance 0.40–0.28, (Hale et al., 2009)).

The results from our trial provides some evidence to support the use of this broader basal area range as a management guide for enhancing bilberry cover in Scots pine plantations. The wider applicability of the prescription requires further field testing. However, this basal area range is consistent with figures for maximum bilberry cover recently reported from an analysis of National Forest Inventory records of Scots pine forest in Finland (Miina et al., 2009) Further, confidence in substituting the prescribed stem density-tree height combinations with the target basal area range can be drawn from the observation that these basal area values provide a transmittance range narrowly centred on transmittance value shown by Parlane et al. (2006) to be optimal for bilberry for Scots pine woodlands in Scotland.

4.3. Compatibility between capercaillie brood habitat enhancement measures and other plantation management objectives

The Parlane et al. (2006) prescription would appear to be compatible with timber production objectives. In the model Scots pine stands these target basal areas were achieved from mid rotation to commercial felling age under a intermediate thinning option. Stands thinned like this therefore would conform to the thinning regime designed to maximise the cumulative volume of timber produced (Rollinson, 1988) and thus offers the possibility of combining timber production with capercaillie conservation (MacMillan and Marshall, 2004). Small patch clearfelling treatment (also considered in our study) offers an alternative method of sourcing timber when increasing intensity of stand thinning is undesirable. It might also be appropriate in managing Scots pine woodlands under shelterwood systems and in the transformation of woodlands from an even-aged to an irregular age structure. Both approaches maintain a continuity of forest cover by avoiding large scale clearfelling at the end of the rotation (Mason et al., 2004).

Clearfell gaps (less than 3 ha in size) in some situations provide the small internal habitat patches and diverse woodland structure which can be an important feature of capercaillie habitat (Gossow and Pseiner, 1981; Kortland, 2006; Watson and Moss, 2008). Further, there is evidence that capercaillie select woodland habitat adjacent to more open forest and that providing a mixture of stands of varying density could meet the range of needs of capercaillie (Gjerde, 1991a). However, more work is needed to ascertain that there are not overriding negative effects of introducing clearfell patches on capercaillie populations, for example through increased habitat fragmentation or predation pressures (King et al., 1998; Chalfoun et al., 2002).

Our work has focussed on plot level effects of thinning even aged and uniform monocultures of Scots pine, and has related the resulting canopy structure (basal area) to response of bilberry. We have not attempted to consider the direct effects on capercaillie of canopy structure at the scale of the site. The dropping count data collected from the plots were too sparse to be informative. This is not surprising as capercaillie territories, particularly in plantations, are big in comparison to the plot sizes used in the trial (Kortland, pers. comm.). Other information, not recorded as part of this study, on capercaille numbers (male birds at leks close to the study sites and brood counts within the study area of Inshriach) (Bibby et al., 2000) do not provide evidence of a response to habitat improvement. Further studies designed to collect such information would be useful in understanding the merits of altering an even plantation canopy through applying a mixture of the treatments recommended here.

Adapting forest management to improve habitat potential for capercillie has been identified as an economical way of optimising conservation efforts at the landscape scale (Braunisch and Suchant, 2007). Recent research shows capercaillie are more flexible in habitat selection than previously assumed (González et al., 2012;

Wegge and Rolstad, 2011), and confirms use of middle-aged plantations (>30 years) by both adults and young broods, as this habitat becomes available (Eaton et al., 2007; Miettinen et al., 2010; Wegge and Rolstad, 2011). Encouraging tree species diversity (to include e.g. larch and spruce) within pine forests may also be important (Gjerde, 1991a; Kortland, 2006). Implementing a silvicultural system where Scots pine plantations are thinned to the improved prescription (which are largely consistent with maximising timber production) and where even age stands, where large, are transformed through thinning and by using small patch clearfells to create multicohort forest structure, is likely to produce desirable outcomes for capercaillie conservation (Gjerde, 1991a; Miettinen et al., 2010). Such a silvicultural system should be considered by managers, particularly in areas where capercaille are present.

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